

MSME Credit Intelligence Platform

How to present an end-to-end data science project in a job interview

A practical walkthrough covering business framing, synthetic banking data, feature engineering, model selection, evaluation, explainability, deployment and the live portal.

Prepared for data science portfolio and interview demonstrations
25 September 2026

01 Your 60-second opening

Begin with the decision problem and business value - not the technology stack.

Use this opening almost word-for-word:

"I built an MSME Credit Intelligence Platform that helps a bank assess small-business borrowers and monitor portfolio risk. I combined customer, account, transaction, credit-bureau, application, loan and repayment data into a point-in-time training dataset. I trained and compared three classification models to predict 12-month default, selected a Random Forest using validation PR-AUC and recall, and deployed it behind FastAPI into an Angular decision portal. The system provides live scoring, Customer 360, transaction analytics, portfolio monitoring, alerts, reports and role-based access."

What this immediately proves

- You understand a real banking decision, not only an algorithm.
- You can work across data engineering, modelling, APIs and product delivery.
- You know how to evaluate an imbalanced classifier and communicate risk trade-offs.
- You can discuss governance honestly because the portfolio is synthetic and the model is an MVP.

Interview rule

Always separate **what the prototype demonstrates** from **what would be required for production**. This makes you sound credible, risk-aware and employable.

02 Business problem and success criteria

Frame the project as a lending decision workflow.

Problem statement

MSMEs often have thin formal credit histories, but their banking activity contains signals about revenue stability, cash-flow pressure and repayment capacity. Manual review is slow and inconsistent. The platform brings these signals together to support faster, more consistent and explainable decisions.

Primary prediction

For a historically approved and observable loan, predict whether it will default within 12 months. In this MVP, the binary target **default_12m** is derived from severe delinquency and loss outcomes.

Stakeholders

Stakeholder	Decision supported	Portal evidence
Credit analyst	Review an application and its risk drivers	Assessment detail, Customer 360
Credit manager	Prioritise referrals and overrides	Assessment queue, recommendations
Risk manager	Monitor concentrations and deterioration	Portfolio risk, alerts, reports
System administrator	Control access and audit activity	Administration and audit logs

Success criteria

- Catch a high proportion of defaults: recall is important because a missed bad loan is costly.
- Control false positives: unnecessary declines lose good customers and revenue.
- Rank risk effectively: ROC-AUC and PR-AUC measure discrimination across thresholds.
- Return calibrated probabilities and decision factors, not only a class label.
- Make the result usable inside a secured workflow with traceability.

03 Data foundation: 13 operational files

Explain the grain of each table before explaining the joins.

Source	Rows	Grain	Purpose
customers.csv	5,000	One row per MSME	Profile, sector, age, turnover, KYC
accounts.csv	10,046	One row per account	Account type, status and balances
account_transactions.csv	400,000	One row per posting	Credits, debits, category, channel, balance
loan_applications.csv	5,500	One row per application	Request, purpose, revenue, expenses, collateral
credit_bureau_reports.csv	5,500	One report per application	Bureau score, facilities, DPD and enquiries
credit_assessments.csv	5,500	Versioned assessment	Score, PD, risk band and recommendation
credit_decisions.csv	5,500	Decision event	Approval/decline, terms and override
loans.csv	3,126	One row per booked loan	Principal, status, DPD, classification
repayment_schedule.csv	54,969	One row per instalment	Due, paid, outstanding and schedule status
loan_payments.csv	40,877	One row per payment	Principal, interest, fees and penalty
risk_alerts.csv	287	One row per alert	Type, severity, status and timestamps
users.csv	4	One row per portal user	Role, branch and account status
audit_logs.csv	1,037	One row per action	Actor, entity, change and timestamp

How to say it

"I designed the synthetic data to resemble normalized bank domains. I first defined primary keys, foreign keys, business constraints and temporal relationships; then generated correlated behaviours so default was learnable but not trivial."

04 Cleaning, joining and point-in-time design

This is where you demonstrate real data science discipline.

Cleaning and validation

- Aligned categorical values with PostgreSQL constraints, for example VERIFIED KYC status and approved transaction categories.
- Validated numeric domains: positive transaction amounts, non-negative balances, scores within range and payment components equal to totals.
- Validated chronology: incorporation before relationship start, disbursement before maturity and resolution after detection.
- Checked uniqueness and referential integrity across customer, application, account, loan and assessment identifiers.
- Profiled missingness and data types before building the training extract. The final training dataset had 54 columns, including the target.

Join strategy

Starting grain	Join or aggregation	Result
Historical booked loan	Loan -> application -> customer	One candidate training row per loan/application
Customer accounts	Aggregate accounts by customer	Account count and oldest relationship
Transactions	Aggregate the 180 days before application	Cash-flow and behavioural features
Bureau report	Join the report associated with the application	External credit behaviour
Loan outcomes	Observe the performance window after booking	default_12m target

Critical point-in-time statement

Features use information available at or before the application date. Loan performance after booking is used only to create the label. This prevents target leakage.

05 Feature engineering

Move from raw banking events to decision-ready signals.

Family	Examples	Interpretation
Business profile	business_age_months, employee_count, annual_turnover_declared	Operating maturity and scale
Affordability	declared_operating_margin, existing_debt_to_revenue	Ability to absorb a new instalment
Request size	requested_amount_to_turnover, requested_term_months	Exposure relative to business capacity
Collateral	collateral_coverage_ratio, collateral_type	Loss mitigation, not willingness to repay
Cash flow	total_credits_180d, total_debits_180d, net_cash_flow_180d	Observed inflows, outflows and surplus
Volatility	transaction_amount_stddev_180d, minimum_balance_180d	Stability and liquidity stress
Behaviour	transaction_count_180d, cash_transaction_count_180d	Activity depth and cash dependence
Bureau	bureau_score, delinquent_facilities, previous_max_dpd_12m	External repayment behaviour
Credit seeking	credit_enquiries_3m, credit_enquiries_6m	Recent demand for credit

Modelling matrix

The final model used **49 predictors**: 41 numeric and 8 categorical. Numeric missing values were imputed; categorical values were imputed and one-hot encoded inside a scikit-learn pipeline so training and live scoring apply identical transformations.

Data quality talking point

One feature, **debit_to_credit_ratio_180d**, had 36 missing values. The missingness was retained and handled in the preprocessing pipeline instead of deleting otherwise useful applications.

06 Model training and selection

Describe the experiment, not just the winning algorithm.

Dataset and split

Dataset	Rows	Defaults	Purpose
Training	1,424	169	Fit candidate models
Validation	475	58	Choose model and threshold
Test	481	56	One-time final evaluation

Candidate comparison on validation

Model	ROC-AUC	PR-AUC	Recall	F1
Random Forest	0.880	0.479	0.810	0.553
Logistic Regression	0.881	0.467	0.810	0.503
Gradient Boosting	0.859	0.441	0.828	0.466

Why Random Forest won

ROC-AUC was almost tied with Logistic Regression, but Random Forest produced the strongest validation PR-AUC and F1 at the required recall level. PR-AUC matters because defaults were the minority class.

Threshold selection

The operating threshold was selected on validation data at **0.364**, targeting roughly 81% recall. It was not selected on the test set. The threshold turns a probability into a workflow action; it is a business-cost decision, not a universal constant.

07 Final evaluation and how to explain it

Use the untouched test set and discuss errors in business language.

0.882

ROC-AUC

0.430

PR-AUC

82.1%

Recall

39.3%

Precision

0.0896

Brier score

Confusion matrix at the selected threshold

	Predicted non-default	Predicted default
Actual non-default	354 true negatives	71 false positives
Actual default	10 false negatives	46 true positives

What to say

- The model correctly identified 46 of 56 defaults, so it captured 82.1% of bad outcomes.
- It missed 10 defaults. In lending, these false negatives are costly because they may become credit losses.
- It flagged 71 non-defaults. Those false positives could lead to unnecessary review or declined good business, so the prediction should support - not replace - credit judgment.
- Precision is moderate because default is uncommon and the threshold intentionally favours recall.
- The Brier score evaluates probability quality; calibration should still be monitored by segment and over time.

Do not say

Do not claim that 88.2% ROC-AUC means the model is 88.2% accurate. AUC measures ranking ability across thresholds; accuracy at the chosen threshold was 83.2%.

08 Customer 360 and live scoring

Explain how the analytical result becomes a usable banking product.

Customer 360 composition

View	Contributing datasets	Examples
Identity and profile	customers	Business type, sector, KYC, turnover, relationship age
Financial behaviour	accounts + account_transactions	Balances, credits, debits, cash flow and activity
Credit demand	loan_applications + credit_decisions	Requests, statuses, decisions and approved terms
Exposure and performance	loans + repayment_schedule + loan_payments	Outstanding principal, arrears, DPD and repayment
Risk intelligence	credit_bureau_reports + credit_assessments + risk_alerts	Bureau history, model PD, risk band and alerts

Live scoring path

PostgreSQL view **credit.ml_scoring_features_v1** recreates the same 49 model inputs for current applications. The validation check found 49 expected features, no missing or extra columns, exact order, 5,500 application rows and a valid scoring probability range.

Example interpretation

Example: PD 61.6% | Risk band VERY HIGH | Internal score 511 | Recommend decline

The portal also shows a recommended limit, term, monthly capacity, debt-service ratio, positive factors, negative factors and policy checks. Bureau score is an external bureau measure; internal credit score is the platform's transformed risk score. They are related but not the same.

09 Your 12-minute live portal walkthrough

Follow one decision story instead of clicking every menu item.

Time	Screen	Narration
0:00-1:00	Login	Mention JWT authentication, bcrypt password hashes, account locking and role-based access.
1:00-2:30	Dashboard	Show portfolio KPIs, risk distribution, alerts and high-risk queue. State what decision each widget supports.
2:30-4:00	Customer 360	Open one MSME. Connect profile, accounts, 180-day cash flow, applications, loans, repayment and risk.
4:00-6:30	Credit assessment	Open a high-risk application. Explain PD, internal score, bureau score, threshold, factors and recommendation.
6:30-7:30	Rescore	Trigger scoring and explain that the live feature view must match the training schema.
7:30-9:00	Transaction Analytics	Filter a customer; show credits, debits, net flow, category patterns and CSV export.
9:00-10:00	Portfolio Risk	Show concentrations and how a risk manager moves from individual decisions to portfolio oversight.
10:00-11:00	Alerts and Reports	Show early-warning follow-up, operational queues and exportable management information.
11:00-12:00	Administration	Explain roles, audit trails and why model decisions require governance.

Demonstration technique

- Choose one borrower story before the interview and know its IDs and expected output.
- Narrate the decision: signal -> model interpretation -> operational action.
- Keep SQL, API documentation and model artifacts open as backup evidence, but do not lead with them.
- If live scoring fails, explain the architecture using saved results instead of debugging silently.
- Finish with limitations and the next production step; this shows maturity.

10 Architecture and engineering talking points

Show that your model can survive outside a notebook.

End-to-end flow

1. Sources	2. PostgreSQL	3. ML pipeline	4. FastAPI	5. Angular
13 CSV domains	Schemas, constraints, feature views	Preprocessing + Random Forest artifact	Scoring, customer and risk services	Role-specific decision portal

What you engineered

- PostgreSQL schemas for banking, credit and security domains with constraints and foreign keys.
- Feature SQL with identical names and ordering for training and live scoring.
- FastAPI layers: endpoints, schemas, services, repositories and database sessions.
- Angular standalone components, routing, HTTP services, interceptors, guards and responsive pages.
- JWT authorization, role-aware navigation, bcrypt password verification and audit logging.
- CSV reporting, transaction analytics and operational risk monitoring.

Why this matters in a data science interview

Many candidates stop at a notebook. This project demonstrates the full lifecycle: defining a target, building reliable features, comparing models, choosing an operating threshold, deploying inference and giving users a governed workflow.

11 Limitations, governance and production roadmap

Your honesty here strengthens the presentation.

Current limitations

- All customer and performance records are synthetic. Metrics demonstrate the pipeline, not expected performance on a real bank portfolio.
- Synthetic relationships can be easier to learn than real economic behaviour and may overstate generalisation.
- The sample has 2,380 training observations and 283 defaults; subgroup estimates may be unstable.
- Feature importance is not causality. Decision factors require robust explainability and policy review.
- No fairness conclusion should be made until protected or proxy groups are defined and tested under the applicable law and bank policy.
- The MVP uses an offline model artifact and local deployment; it is not yet a highly available production scoring service.

Production roadmap

Priority	Next step	Why
1	Replace synthetic data with governed historical snapshots	Realistic performance and operational validity
2	Temporal/out-of-time validation and backtesting	Measure stability across economic periods
3	Calibration, segment and fairness testing	Reliable, defensible probabilities
4	Model registry, approval workflow and versioned features	Reproducibility and governance
5	Drift, performance and override monitoring	Detect deterioration after deployment
6	Docker, CI/CD, secrets, migrations and observability	Secure and maintainable operations

12 Likely interview questions and strong answers

Keep answers specific to what you built.

Question	Suggested answer
Why not use accuracy?	Defaults are a minority. Accuracy can remain high while missing risky borrowers. I used ROC-AUC and PR-AUC for ranking, recall for default capture, precision and F1 for threshold behaviour, and Brier score for probability quality.
Why Random Forest over Logistic Regression?	Their ROC-AUC values were almost tied, but Random Forest achieved higher validation PR-AUC and F1 at the target recall. Logistic Regression remains a useful interpretable challenger.
How did you avoid leakage?	Features were calculated from information available by the application date, including a backward-looking 180-day transaction window. Post-booking performance was used only for the target.
Why is the threshold 0.364?	It was chosen on validation data to achieve approximately 81% recall while improving F1. In production I would select it using quantified costs, capacity and risk appetite.
What would you monitor?	Input drift, missingness, unseen categories, score distribution, calibration, default rate, recall/precision when labels mature, overrides, segment performance and service latency.
How do you explain a score?	I show the PD, risk band, threshold, external bureau evidence, affordability, cash-flow signals and policy checks. I avoid treating feature importance as causality.
What was the hardest part?	Maintaining consistent grains and point-in-time joins across transactions, applications, bureau reports and loan outcomes, then keeping the live feature view compatible with the training pipeline.

13 How to position the project in your job search

Turn the build into evidence recruiters can scan quickly.

Resume bullets

- Built an end-to-end MSME credit-risk platform using PostgreSQL, Python/scikit-learn, FastAPI and Angular, integrating 13 banking data domains and 400,000 transactions.
- Engineered 49 point-in-time credit-risk features across cash flow, affordability, bureau history and collateral; created a 2,380-row historical training cohort with an 11.9% default rate.
- Compared Logistic Regression, Random Forest and Gradient Boosting; selected a Random Forest achieving 0.882 test ROC-AUC, 0.430 PR-AUC and 82.1% default recall.
- Deployed live model scoring, Customer 360, transaction analytics, portfolio monitoring, alerts, reporting, JWT authentication and role-based access controls.

Portfolio repository checklist

- README with business problem, architecture, dataset grains and screenshots.
- Reproducible data-generation and loading commands; never commit passwords or real customer data.
- Training script, comparison results, threshold rationale and model card.
- API examples, database diagram and a short demo video.
- A limitations section that clearly labels the data and metrics as synthetic-MVP evidence.

Closing statement

"This project shows how I approach data science as a decision system: I start from the business cost, create trustworthy point-in-time data, evaluate models with the right metrics, deploy the result into a usable workflow, and plan for governance and monitoring."

Final preparation

Practise the 60-second opening, the 12-minute demo and three metric explanations aloud. Record yourself once. Remove any step that sounds like a feature tour rather than a decision story.